STA 610 Case Study 2

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1 Introduction and Background

In the 2020 United States presidential election, Democratic candidate Joe Biden faced Republican incumbent Donald Trump, with a notable shift in voting methods due to the COVID-19 pandemic. North Carolina's 15 electoral votes contributed to Trump's overall tally of 232, while Biden received 306 electoral votes [1]. I retrieved the North Carolina election results data from the North Carolina State Board of Elections (NCSBE).

The data consists of unaggregated historical voter registration data and voter results data from the 2020 General Election. From the NSCBE, *"voter stats files contain registered voter demographics at the time of a given election in North Carolina; these voters did not necessarily cast a vote in the election."* The data is at the county and precinct level, and demographic attributes include age (expressed as a range, i.e. 41-65), political party, race, ethnicity, and sex.

2 EDA

2.1 Data Cleaning and Preparation

The instructions note that *"unfortunately, the data dictionary from the NCSBE does not provide the exact difference between the variables party_cd and voted_party_cd"*. Since I could not find a better answer online, I emailed the NCSBE. Colin Loftin, an Elections Data Analyst at the NCSBE replied and said that *The party_cd column represents the registered party of the particular voter at the time the file was generated while voted_party_cd represents the party's ballot the voter chose to vote in the given election.* Based on this distinction, I used party_cd.

Since the data is at the precinct level (a subdivision of counties where individuals are assigned to vote), I summarized the voter counts and combined the registration data with the votes cast data.

Next, I checked for any data quality issues. When the data was joined, I calculated the voter turnout as (# of Votes/# Registered Voters). However, some subsets have proportions greater than 1, indicating there were more ballots cast than registered voters. The cause of this is unknown. Since it only appears for 11 observations of 51,906 total, which equates to 55 cast votes, I feel comfortable removing them from the data. I also manually adjusted the names for the political parties for easier interpretation. Since the U.S. has a two-party system, I grouped the parties into "Democrat", "Republican", and "Other Party". The data dictionary was also missing a description for the race code "P", which stands for Pacific Islander[2]. Last, when the data was joined, there were 4,099 NA values for actual voters, indicating that there was a record of voters being registered that didn't vote for a given demographic combination. In most cases, these records only had 1 or 2 registered voters. I filled the actual_voter values with 0 for these records. After these adjustments were made, the data was joined on all the relevant demographic characteristics at the county level.

2.2 County Sampling

Our instructions say to sample 30 counties. However, we don't want a truly random sample, or we could get very unlucky and get 30 counties with a very small number of voters, or voters that don't represent the overall population very well. Using techniques I learned in STA 522 (Study Design), I used a probability proportional to size sampling technique. In this approach, observations are weighted based on some variable (in my case I used the number of registered voters), then sampled based on those weights. In survey terminology, each county has a first-order inclusion probability π_j. In a simple random sample, this is $\frac{n}{N}$. If we used that for the counties, we would have $\pi_j = 30/100 = 0.3$ for every county. With our pps method though, we have $\pi_j^{pps} = \frac{nM_j}{\sum_{j=1}^{N} M_j}$, where $n = 30$, M_j is the numbered of registered voters in county j, and $\sum M_j$ is the total number of registered voters in NC in 2020. Also note that because pps sampling draws with replacement, I had to sample a few more than 30 to get 30 unique counties. This resulted in the following counties: Ashe, Brunswick, Buncombe, Cabarrus, Catawba, Cleveland, Cumberland, Currituck, Duplin, Durham, Forsyth, Franklin, Granville, Guilford, Henderson, Iredell, Johnston, Macon, Mecklenburg, Nash, New Hanover, Orange, Pitt, Randolph, Rowan, Stanly, Union, Wake, Watauga, and Yadkin. Considering

the large cities in North Carolina, Charlotte is in Mecklenburg county, Winston-Salem is in Forsyth county, and Raleigh is in Wake county, but several of the other counties are more rural. These counties are visualized in Figure 6.

2.3 EDA Considerations

To start thinking about how I wanted to build a model, I looked at voter turnout across the counties and among different demographic characteristics. Considering the turnout rates across different demographic combinations in Figure 1, the striking difference is how age affects turnout rates. This looks to me like it will require an interaction effect in our model for age and party affiliation. Age in general is associated with higher turnout rates, which is unsurprising. White individuals vote at slightly higher rates than other races at around 0.7, while Black individuals have the lowest turnout rates at 0.6. Pacific Islanders voted at a very high rate compared to the rest of the population, though my sampled counties only contained 311 Pacific Islander registered voters. Last, Non-Hispanic or Latino individuals vote at higher rates than Hispanic or Latino individuals by about 10%.

Figure 2 shows the differences in turnout by county across all demographic combinations. There are some noticeable differences between the means of, say, Pitt County and Wake County, so this points towards a county-level random intercept.

Last, I tried to visualize whether or not fitting random slopes would make sense. Based on the plots in 3, it does not seem initially convincing that random slopes are appropriate.

3 Modeling

3.1 Final Model

I modeled the turnout rates with a logistic GLMM as follows for i units and j counties. Note that each unit i is registration and vote data summarized at the county level for a unique demographic slice, so I'm thinking about votes out of registered voters as the number of successes out of N trials. To capture differences across counties, we'll use a random intercept.

$$
\begin{aligned}
y_{ij}|x_{ij} &\sim \text{BIN}(\text{num_registered}_{ij},\ \text{turnout}_{ij}) \\
\text{logit}(\text{turnout}_{ij}) &= (\beta_0 + b_{0j}) + \beta_1\text{party}_{ij} + \beta_2\text{race}_{ij} + \beta_3\text{age}_{ij} + \beta_4\text{ethnicity}_{ij} \\
&\quad + \beta_5\text{sex}_{ij} + \beta_6\text{party*sex}_{ij} + \beta_7\text{party*age}_{ij} \\
b_{0j} &\sim \mathcal{N}(0, \sigma^2), \quad j = 1, \ldots, 30, i = 1, \ldots, n
\end{aligned}
$$

3.2 Modeling Process

Since we're interested in every predictor, I didn't perform any variable selection and started by fitting a model with a random intercept at the county level based on EDA along with every covariate. Next, I wanted to incorporate something to help answer questions 3 and 4 in the prompt about turnout rates differing between sex and party affiliation, and turnout rates between age groups for different party affiliations. This sounds like great place to use interaction terms, but I wanted to try random slopes just in case. However, these models didn't converge, so I stuck with the interaction terms. I also tried a model with random intercepts for the interaction terms, though that model resulted in a similar deviance as the fixed effects model. For simplicity of interpretation, I kept the fixed-effect interaction model, though this has the drawback of decreased degrees of freedom. Though note that a significant limitation of the model is a lack of generalizability to other states that have very different demographic characteristics.

3.3 Model Fit

We have relatively fewer tools to validate a model like this compared to, say, a linear model. Because we aren't making an assumption about the distribution of the errors, it doesn't make sense to try to examine residuals for evidence of normality. However, we can look at the deviance residuals which are presented

in Figure 5. There is no visible structure and they are centered at zero, which indicates the model fits reasonably well.

4 Results and Conclusions

The model estimated $\hat{\sigma} = 0.19$, with a 95% C.I. of $[0.14, 0.23]$, and the estimates, standard errors, and confidence intervals for both the county-level random intercepts and fixed effects are in Table 2 and Table 1 respectively. Also note that the baseline factors are party = Democrat, sex = Female, age = Age 18-25, race = Asian, and ethnicity = Hispanic.

4.1 Questions of Interest

How did different demographic subgroups vote in the 2020 general elections?

This question is partially answered by our EDA and Figure 1, but we can add some detail from the coefficients in Table 1. Compared to Democrats, individuals that identify as Republican or Other parties have decreased odds of voting. Males have 0.755x the odds of voting compared to females, but unidentified sex individuals vote at a higher rate of 1.314x the odds of the baseline. Again, we see that older individuals vote at significantly higher rates than younger, with Age 41 - 65 at 2.533x the baseline of Age 18-25 and Age 66+ at 2.648x the baseline. Pacific Islanders vote at 4.347x the odds of Asian individuals, the baseline in that category, while both White and Undesignated Races vote at about 1.21x the baseline. Last, Hispanic or Latino individuals vote at lower rates than undesignated ethnicities or Not Hispanic or Latino individuals. None of the confidence intervals cover 0 as well, so that is reassuring for our estimates. This does not include an interpretation of every coefficient, but the rest follow a similar logic of a multiplicative change in odds over the baseline.

Did the overall probability or odds of voting differ by county in 2020? Which counties differ the most from other counties?

Considering the dotplot in Figure 4, there is clear evidence of heterogeneity in turnout rates across counties, so the log odds differ by county. In particular, Wake county had significantly higher log odds of voter turnout (the dotplot is not exponentiated), while Cumberland county had significantly lower log odds than the other counties. None of the confidence intervals cover 0, so I am reasonably confident that there is heterogeneity. Looking at Table 2, Wake county has a 1.38x increase in the odds of voter turnout (looking at the exponentiated coefficients), while Cumberland county, which we identified above, has 0.695 the odds of turnout.

How did the turnout rates differ between females and males for the different party affiliations?

The baseline for the party : sex_code interaction is Democrat Female. Interpreting the odds in Table 1, the turnout for Men is higher than Democrat Women for Other and Republican parties. Because we just have one baseline level, we can interpret the Democrat Men effect as the single fixed estimate for sex_code = M since Democrat is the baseline party level. This tells us the turnout for Democrat Men is 0.755x the baseline level, indicating that they vote at lower levels than Democrat Females. Last, Republican Women vote at 0.85x the baseline level and Women who register in an "Other Party" vote at 0.61x the baseline level, both using similar logic.

How did the turnout rates differ between age groups for the different party affiliations?

The party:age interaction baseline is Democrat Ages 18 - 25. In general, this interaction effect looks more similar across combinations with most estimates slightly above 1, except individuals Age 66 + : Other party, which is 1.387x the baseline. Since we saw that younger people vote at lower rates, these interactions make sense. Democrats Age 26-40 vote at 1.051x the baseline, Democrats Age 41-65 vote at 2.533x the baseline, and Democrats Age 66+ vote at 2.649x the baseline. All of these come from reading the main effects since Democrat is the baseline.

References

[1] Allan James Vestal et. al. *Joe Biden has been declared the winner, toppling Donald Trump after four years of upheaval in the White House.* Accessed: December 3, 2023. 2023. URL: https://www.politico.com/2020-election/results/president/.

[2] Department of Children and State of Kansas Families. *Instructions for Reporting Race, Ethnicity, Language Preference and Other Communication Media.* Accessed: December 3, 2023. 2023. URL: https://www.dcf.ks.gov/services/PPS/Documents/PPM_Forms/Section_5000_Forms/PPS5460_Instr.pdf.

5 Appendix

Table 1: Fixed Effects

	exp{estimate}	estimate	std error	2.5%	97.5%
(Intercept)	1.407	0.341	0.026	0.315	0.368
partyOther Party	0.606	−0.502	0.007	−0.508	−0.495
partyRepublican	0.850	−0.163	0.008	−0.171	−0.155
sex_codeM	0.755	−0.281	0.004	−0.285	−0.278
sex_codeU	1.314	0.273	0.009	0.264	0.282
ageAge 26 - 40	1.051	0.050	0.005	0.045	0.056
ageAge 41 - 65	2.533	0.929	0.006	0.924	0.935
ageAge Over 66	2.649	0.974	0.006	0.968	0.981
raceBlack	0.742	−0.298	0.008	−0.306	−0.290
raceNative American	0.730	−0.315	0.020	−0.335	−0.294
raceOther Race	0.775	−0.254	0.010	−0.264	−0.245
racePacific Islander	4.347	1.469	0.210	1.259	1.680
racetwo_or_more	0.812	−0.208	0.014	−0.221	−0.194
raceUndesig. Race	1.219	0.198	0.009	0.188	0.207
raceWhite	1.224	0.202	0.008	0.195	0.210
ethnicityNot Hispanic Latino	1.542	0.433	0.006	0.427	0.440
ethnicityUndesig. Eth.	1.370	0.315	0.006	0.308	0.321
partyOther Party:sex_codeM	1.173	0.160	0.005	0.155	0.165
partyRepublican:sex_codeM	1.275	0.243	0.006	0.237	0.249
partyOther Party:sex_codeU	0.796	−0.229	0.010	−0.238	−0.219
partyRepublican:sex_codeU	1.283	0.250	0.013	0.237	0.262
partyOther Party:ageAge 26 - 40	1.157	0.146	0.007	0.139	0.153
partyRepublican:ageAge 26 - 40	1.064	0.062	0.009	0.053	0.071
partyOther Party:ageAge 41 - 65	1.093	0.089	0.007	0.082	0.096
partyRepublican:ageAge 41 - 65	1.089	0.085	0.009	0.076	0.094
partyOther Party:ageAge Over 66	1.387	0.327	0.009	0.318	0.336
partyRepublican:ageAge Over 66	1.082	0.079	0.010	0.069	0.089

Table 2: Random Intercepts by County

	County	exp{estimate}	estimate	std_dev	2.5%	97.5%
1	ASHE	1.083	0.079	0.019	0.043	0.116
2	BRUNSWICK	0.928	−0.074	0.007	−0.089	−0.060
3	BUNCOMBE	1.076	0.073	0.005	0.062	0.084
4	CABARRUS	1.164	0.152	0.006	0.140	0.165
5	CATAWBA	0.965	−0.036	0.007	−0.050	−0.021
6	CLEVELAND	0.924	−0.079	0.009	−0.097	−0.061
7	CUMBERLAND	0.695	−0.363	0.005	−0.372	−0.354
8	CURRITUCK	0.844	−0.169	0.016	−0.200	−0.138
9	DUPLIN	0.863	−0.148	0.013	−0.174	−0.121
10	DURHAM	1.089	0.085	0.005	0.076	0.094
11	FORSYTH	0.958	−0.043	0.005	−0.052	−0.034
12	FRANKLIN	1.151	0.141	0.011	0.118	0.163
13	GRANVILLE	1.133	0.125	0.012	0.101	0.150
14	GUILFORD	1.021	0.020	0.004	0.013	0.028
15	HENDERSON	0.976	−0.024	0.008	−0.040	−0.007
16	IREDELL	1.082	0.079	0.007	0.066	0.093
17	JOHNSTON	1.112	0.106	0.007	0.093	0.119
18	MACON	0.831	−0.185	0.015	−0.214	−0.156
19	MECKLENBURG	0.942	−0.060	0.003	−0.065	−0.055
20	NASH	1.054	0.053	0.009	0.035	0.071
21	NEW HANOVER	0.900	−0.105	0.006	−0.116	−0.094
22	ORANGE	1.085	0.081	0.007	0.067	0.095
23	PITT	0.891	−0.116	0.006	−0.128	−0.103
24	RANDOLPH	0.975	−0.025	0.008	−0.041	−0.010
25	ROWAN	0.924	−0.079	0.008	−0.094	−0.064
26	STANLY	1.079	0.076	0.012	0.052	0.099
27	UNION	1.102	0.097	0.006	0.085	0.109
28	WAKE	1.380	0.322	0.003	0.316	0.328
29	WATAUGA	0.891	−0.115	0.011	−0.136	−0.094
30	YADKIN	1.139	0.130	0.016	0.098	0.162

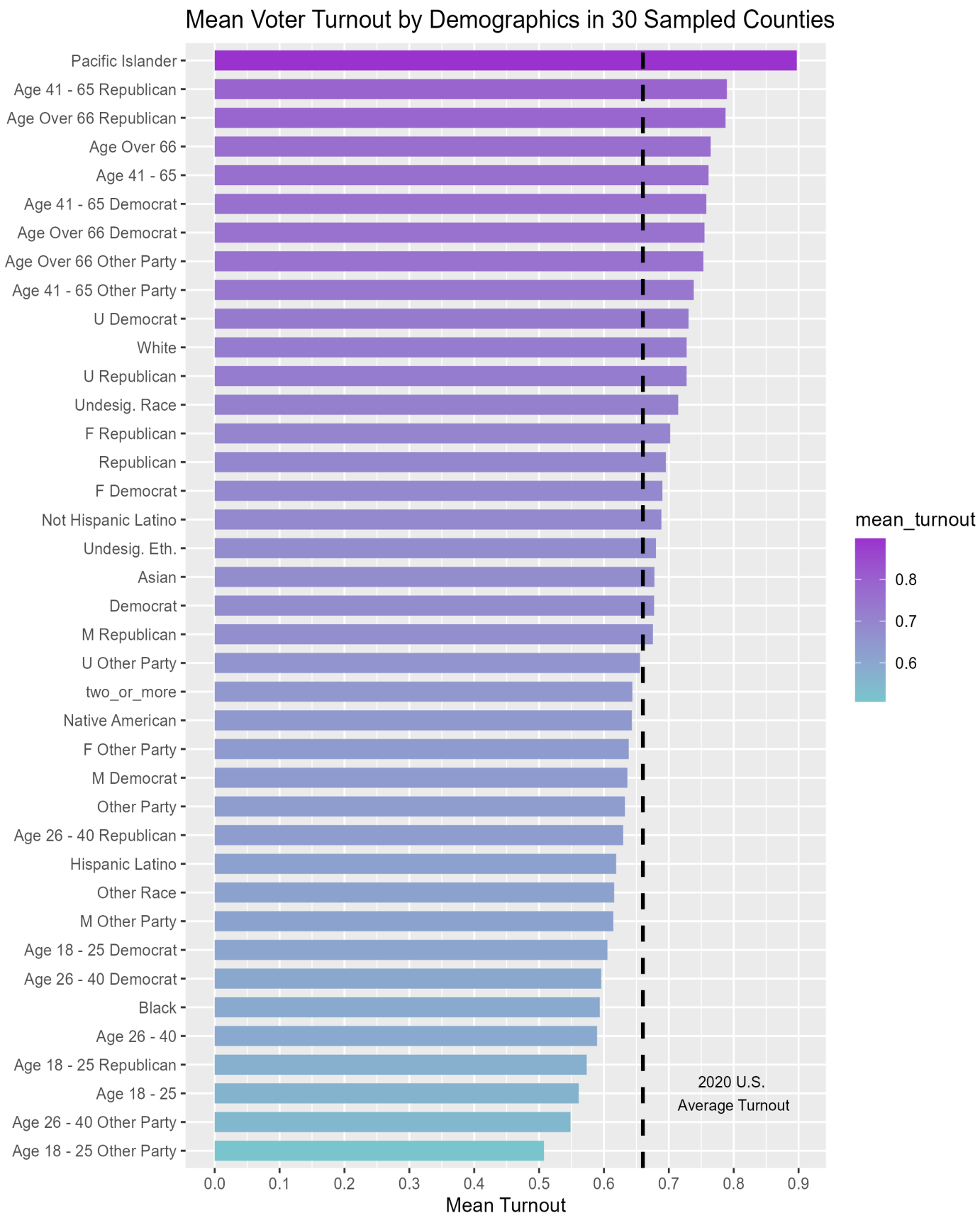

Figure 1: Voter Turnout by Demographics

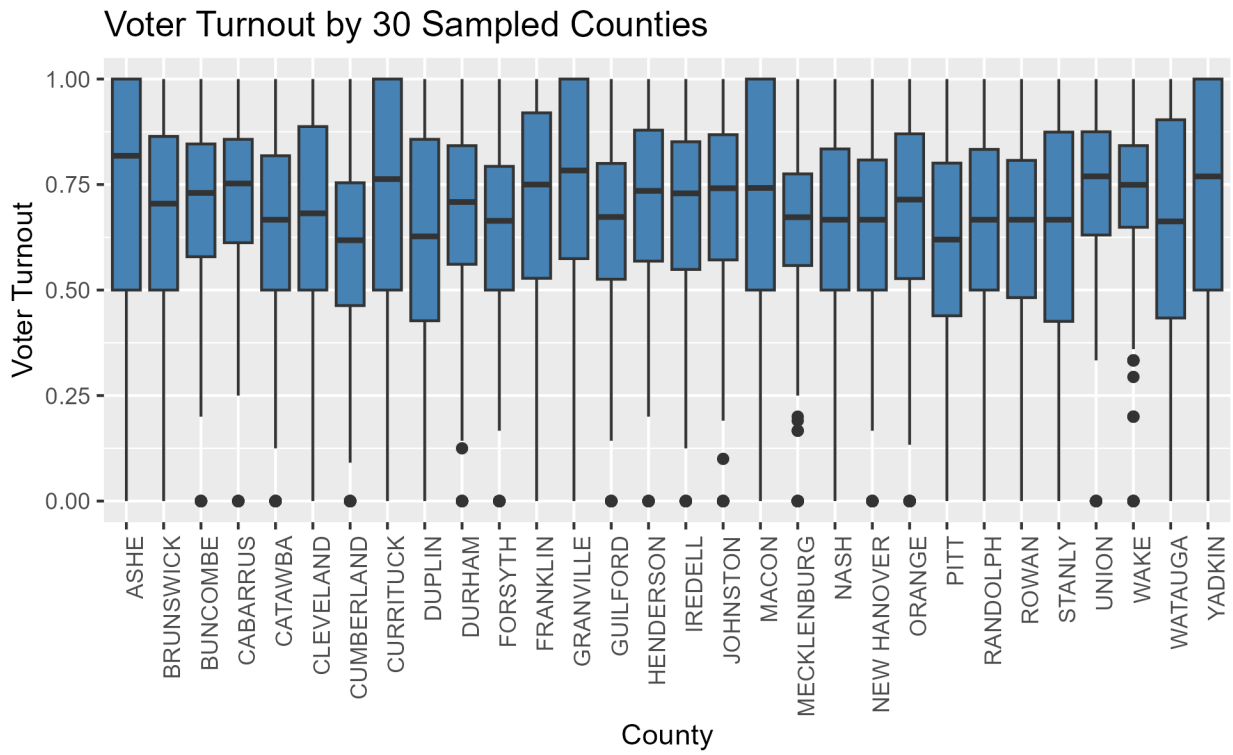

Figure 2: Voter Turnout by County

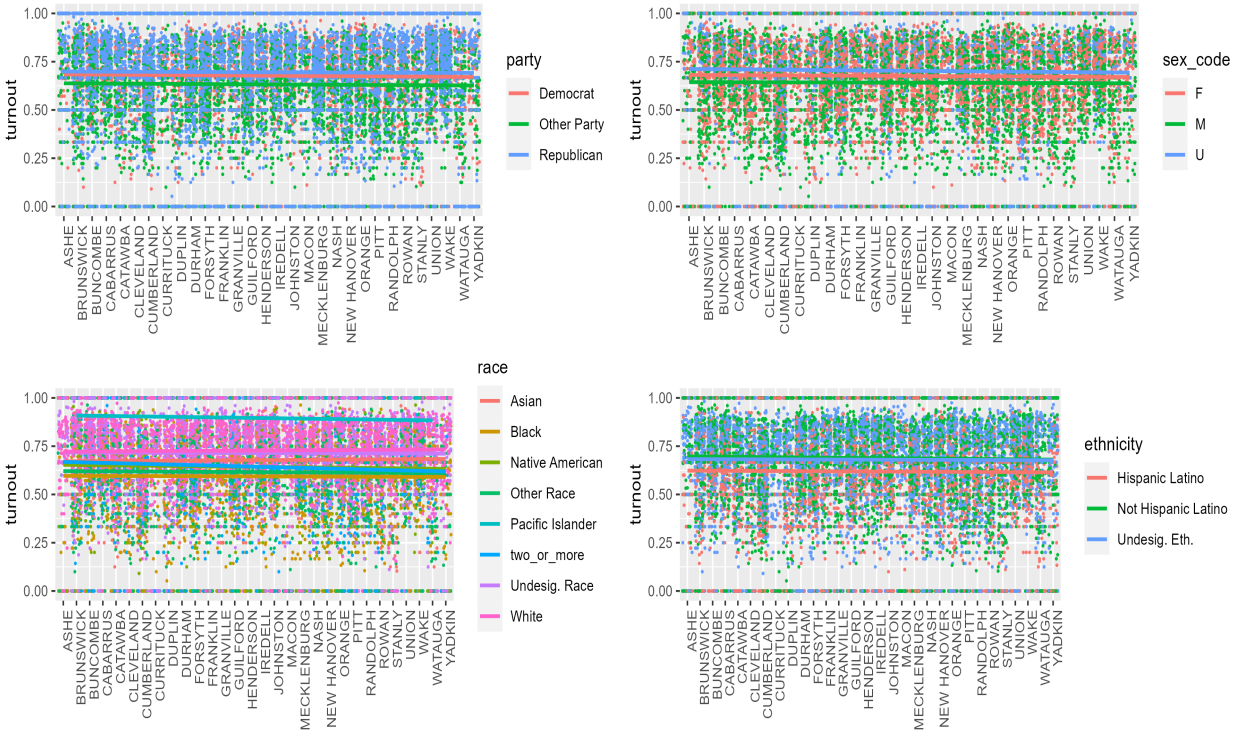

Figure 3: Random Slopes by Demographics

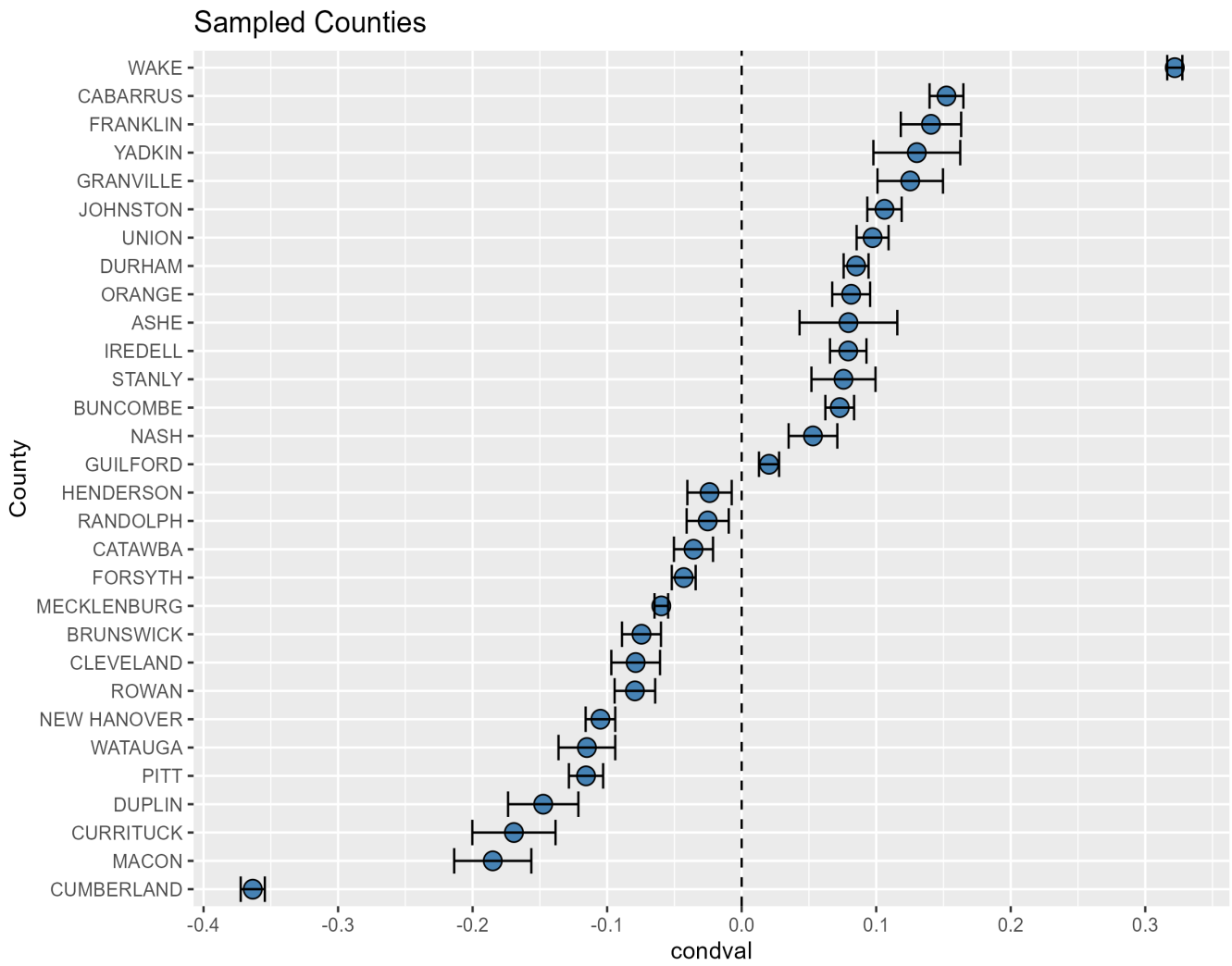

Figure 4: Random Effects by County

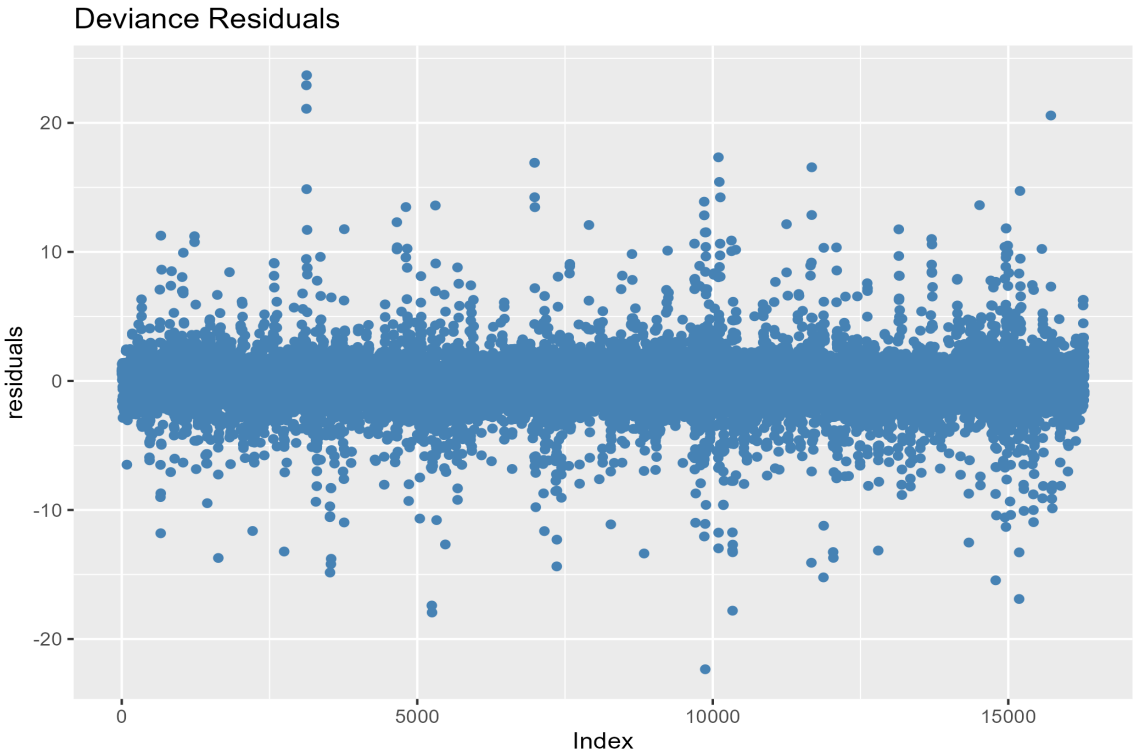

Figure 5: Deviance Residuals

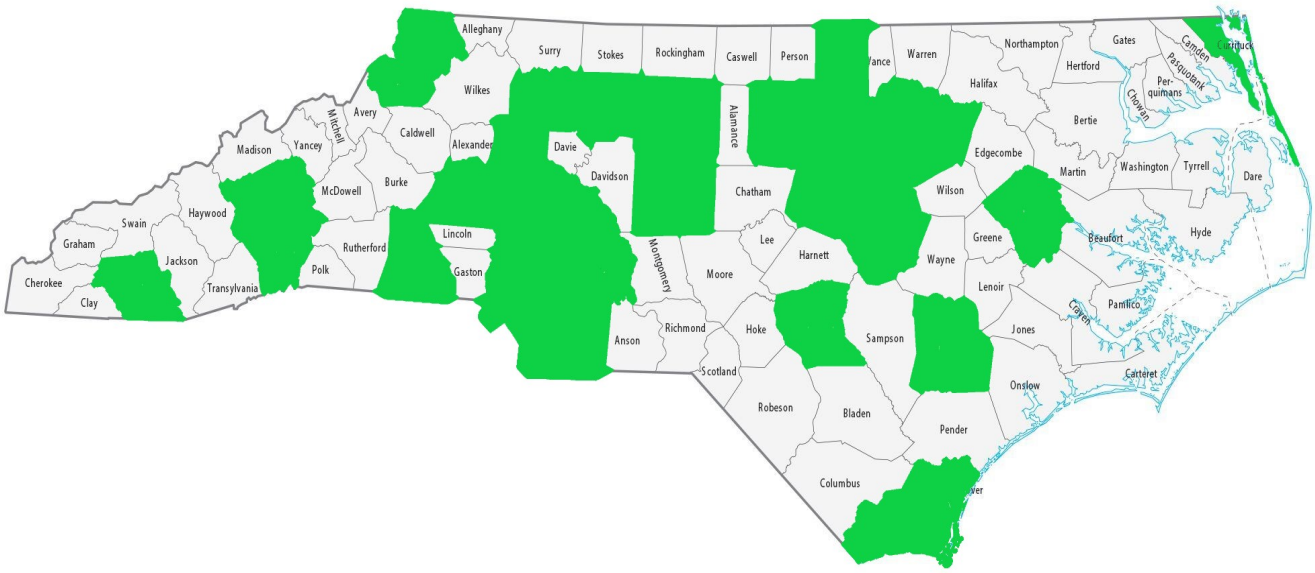

Figure 6: Sampled N.C. Counties

6 Code Appendix

```
---
title: "610 Case Study 2"
format: pdf
---

# Load

```{r, include=FALSE}
library(tidyverse)
library(lme4)
library(pps)
library(cowplot)
library(stargazer)

vote = read.csv('voter_stats_20201103.txt', sep = "\t")
history = read.csv('history_stats_20201103.txt', sep = "\t")
```

1. Cleaning

```{r}

# history is actual voters
history = history |>
  mutate(party = case_when(party_cd == "DEM" ~ 'Democrat',
                           party_cd == "REP" ~ 'Republican',
                           TRUE ~ "Other Party")) |>
  group_by(county_desc,
           party,
           race_code,
           ethnic_code,
           sex_code,
           age) |>
  summarize(actual_voters = sum(total_voters))

# registered voters
vote = vote |>
  mutate(party = case_when(party_cd == "DEM" ~ 'Democrat',
                           party_cd == "REP" ~ 'Republican',
                           TRUE ~ "Other Party")) |>
  group_by(
    county_desc,
    party,
    race_code,
    ethnic_code,
    sex_code,
    age
  ) |>
  summarize(reg_voters = sum(total_voters))


total_voting = left_join(
```
```

```
  vote,
  history,
  by = c("county_desc", "party", "race_code", "ethnic_code",
         "sex_code", "age")
  ) |>
  mutate(race = case_when(race_code == "A" ~ "Asian",
                          race_code == "B" ~ "Black",
                          race_code == "I" ~ "Native American",
                          race_code == "M" ~ "two_or_more",
                          race_code == "O" ~ "Other Race",
                          race_code == "W" ~ "White",
                          race_code == "P" ~ "Pacific Islander",
                          TRUE ~ "Undesig. Race"),
         ethnicity = case_when(ethnic_code == "HL" ~ 'Hispanic Latino',
                               ethnic_code == "NL" ~ 'Not Hispanic Latino',
                               ethnic_code == "UN" ~ 'Undesig. Eth.'),
         actual_voters = replace_na(actual_voters, 0),
         turnout = actual_voters / reg_voters) |>
  ungroup() |>
  select(-c(race_code, ethnic_code)) |>
  filter(turnout <= 1)
```

2. Sample the counties

```{r}
set.seed(1780)

# first aggregate everything so we can pps sample
aggregated = total_voting |>
  group_by(county_desc) |>
  summarize(total_registered = sum(reg_voters)) |>
  ungroup()

# draw 37 samples with replacement
county_index = ppss(aggregated$total_registered, 37)

# print county list
samp_counties = unique(aggregated[county_index, ]$county_desc)
length(samp_counties)
print(samp_counties)

sampled_counties = total_voting |>
  filter(county_desc %in% samp_counties)
```

3. Plots

```{r}
# First aggregate data together to plot later.
agg_char = bind_rows(

  sampled_counties |>
```

```
    group_by(race) |>
    summarize(mean_turnout = mean(turnout)) |>
    mutate(id = race) |>
    select(id, mean_turnout),

  sampled_counties |>
    group_by(party) |>
    summarize(mean_turnout = mean(turnout)) |>
    mutate(id = party) |>
    ungroup() |>
    select(id, mean_turnout),

  sampled_counties |>
    group_by(age) |>
    summarize(mean_turnout = mean(turnout)) |>
    mutate(id = age) |>
    ungroup() |>
    select(id, mean_turnout),

  sampled_counties |>
    group_by(ethnicity) |>
    summarize(mean_turnout = mean(turnout)) |>
    mutate(id = ethnicity) |>
    ungroup() |>
    select(id, mean_turnout),

  sampled_counties |>
    group_by(sex_code, party) |>
    summarize(mean_turnout = mean(turnout)) |>
    mutate(id = paste(sex_code, party)) |>
    ungroup() |>
    select(id, mean_turnout),

  sampled_counties |>
    group_by(age, party) |>
    summarize(mean_turnout = mean(turnout)) |>
    mutate(id = paste(age, party)) |>
    ungroup() |>
    select(id, mean_turnout)
  ) |>
  mutate(id = fct_reorder(id, mean_turnout))
```

Plot 1

```{r}
agg_char |>
  ggplot(aes(x = mean_turnout, y = id, fill=mean_turnout)) +
  geom_col(width = 0.7) +
  scale_fill_gradient(low = "cadetblue3", high = "darkorchid3") +
  scale_x_continuous(breaks = seq(0, 1, by = 0.1)) +
  geom_vline(xintercept = 0.66, linewidth = 1, linetype = 'dashed') +
  labs(
    title = "Mean Voter Turnout by Demographics in 30 Sampled Counties",
```

```
    y = "",
    x = "Mean Turnout"
  ) +
    annotate(
    "text",
    x = 0.8,
    y = 3,
    label = "2020 U.S. \nAverage Turnout",
    color = "black",
    size = 3,
    vjust = 0.5  # Adjust vertical position
  )

ggsave("images/voter_turnout_demographics.png",
       plot = last_plot(), width = 8, height = 9, units = 'in')
```


```{r}
# Boxplot for question 2

sampled_counties |>
  ggplot(aes(x = county_desc, y = turnout)) +
  geom_boxplot(fill = 'steelblue') +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Voter Turnout by 30 Sampled Counties",
    x = "County",
    y = "Voter Turnout"
  )

ggsave("images/voter_turnout_county.png",
       plot = last_plot(), width = 6.5, height = 4, units = 'in')
```


```{r}
# Checking for random slopes

# random slope for party?
p1 = sampled_counties |>
  ggplot(aes(x = county_desc, y = turnout, color = party)) +
  geom_jitter(size = 0.3) +
  geom_smooth(method = "lm", se = FALSE, aes(group = party)) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(x = "")

# random slope for sex?
p2 = sampled_counties |>
  ggplot(aes(x = county_desc, y = turnout, color = sex_code)) +
  geom_jitter(size = 0.3) +
  geom_smooth(method = "lm", se = FALSE, aes(group = sex_code)) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(x = "")
```

```
# random slope for sex?
p3 = sampled_counties |>
  ggplot(aes(x = county_desc, y = turnout, color = race)) +
  geom_jitter(size = 0.3) +
  geom_smooth(method = "lm", se = FALSE, aes(group = race)) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(x = "")

# random slope for ethnicity?
p4 = sampled_counties |>
  ggplot(aes(x = county_desc, y = turnout, color = ethnicity)) +
  geom_jitter(size = 0.3) +
  geom_smooth(method = "lm", se = FALSE, aes(group = ethnicity)) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(x = "")

combined_plots = plot_grid(p1, p2, p3, p4)

ggsave("images/random_slopes.png",
       plot = last_plot(), width = 12, height = 6, units = 'in')
```

4. Modeling

```{r}

# function to make dotplots later
custom_dotplot = function(model) {

  ranef_data = data.frame(ranef(model))

  ranef_data |>
    ggplot(aes(x = condval, y = grp)) +
    geom_dotplot(binaxis = 'y', stackdir = 'center',
                 dotsize = 0.7, fill='steelblue') +
    geom_errorbar(data = ranef_data,
                  aes(x = condval, y = grp,
                      xmin = condval - 1.96 * condsd,
                      xmax = condval + 1.96 * condsd)) +
    labs(
      title = "Sampled Counties",
      y = "County") +
    scale_x_continuous(breaks = seq(-1, 1, by = 0.1)) +
    geom_vline(xintercept = 0, linetype = 'dashed')

}
```

4.1) GLMM Models

```{r}
m1 = glmer(cbind(actual_voters, reg_voters - actual_voters) ~ (1 | county_desc) +
           party + sex_code + age + race + ethnicity,
```

```
            data = sampled_counties,
            family = binomial)

summary(m1)
custom_dotplot(m1)
```

Have to calculate confidence interval this way, full model takes too long to run

```{r}
m_test = glmer(cbind(actual_voters, reg_voters - actual_voters) ~
                  (1 | county_desc),
            data = sampled_counties,
            family = binomial)

confint(m_test, oldNames = FALSE)
```

4.1.1) Final Model

```{r}
m2 = glmer(cbind(actual_voters, reg_voters - actual_voters) ~ (1 | county_desc) +
            party*sex_code + party*age + race + ethnicity,
            data = sampled_counties,
            family = binomial)

final_model = m2

summary(m2)
```

Validation

```{r}
resids = data.frame(resid(m2, type = 'deviance')) |>
  mutate(y = seq(1,16287))
colnames(resids) = c('residuals', 'y')

resids |>
  ggplot() +
  geom_point(aes(x = y, y = residuals), color= 'steelblue') +
  labs(
    title = 'Deviance Residuals',
    x = 'Index'
  )

ggsave("images/residuals.png",
       plot = last_plot(), width = 7, height = 5, units = 'in')
```

Tried random slopes by county, fails to converge

```{r}
m3 = glmer(cbind(actual_voters, reg_voters - actual_voters) ~ (1 | county_desc) +
```

```
            (party | county_desc) + party*sex_code + party*age + race + ethnicity,
            data = sampled_counties,
            family = binomial)

m4 = glmer(cbind(actual_voters, reg_voters - actual_voters) ~ (1 | county_desc) +
            (age | county_desc) + party*sex_code + party*age + race + ethnicity,
            data = sampled_counties,
            family = binomial)

summary(m3)
summary(m4)
```

Random intercepts, similar AIC as interaction model

```{r}
m5 = glmer(cbind(actual_voters, reg_voters - actual_voters) ~ (1 | county_desc) +
            (1 | party:sex_code) + (1 | party:age) + race + ethnicity,
            data = sampled_counties,
            family = binomial)

summary(m5)
```

5. Getting tables out

```{r}
custom_dotplot(final_model)

ggsave("images/dotplot.png",
       plot = last_plot(), width = 8, height = 6, units = 'in')
```

Fixed effects table

```{r}
confint_table = data.frame(summary(final_model)$coefficients) |>
  select(Estimate, Std..Error) |>
  mutate(exp_beta = exp(Estimate),
         `lb_2.5%` = Estimate - Std..Error,
         `ub_97.5%` = Estimate + Std..Error) |>
  select(exp_beta, Estimate, Std..Error, `lb_2.5%`, `ub_97.5%`)

stargazer(confint_table, summary = FALSE)
```

Random effects table

```{r}
ranef_coeffs = data.frame(ranef(final_model)) |>
  select(grp, condval, condsd) |>
  rename(estimate = condval,
         std_dev = condsd) |>
  mutate(exp_beta = exp(estimate),
```

```
         `lb_2.5%` = estimate - 1.96 * std_dev,
         `ub_97.5%` = estimate + 1.96 * std_dev) |>
  select(grp, exp_beta, estimate, std_dev, `lb_2.5%`, `ub_97.5%`)

stargazer(ranef_coeffs, summary = FALSE)
```